$$\overrightarrow{OG_2} = \frac{1\vec{b} + 2\left(\frac{\vec{c}+\vec{a}}{2}\right)}{3} = \frac{\vec{a}+\vec{b}+\vec{c}}{3}. \quad (2)$$

Similarly if G_3 divides CF in the ratio 2 : 1 then

$$\overrightarrow{OG_3} = \frac{\vec{a}+\vec{b}+\vec{c}}{3} \quad (3)$$

From (1), (2), and (3) we find that the position vectors of the three points G_1, G_2, G_3 are one and the same. Hence they are not different points. Let the common point be G.

Therefore the three medians are concurrent and the point of concurrence is G.

Theorem 8.4

A quadrilateral is a parallelogram if and only if its diagonals bisect each other.

Proof

Let A, B, C, D be the vertices of a quadrilateral with diagonals AC and BD. Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be the position vectors of A, B, C, and D respectively with respect to O.

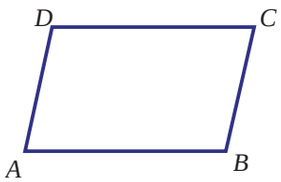


Fig. 8.28

Let the quadrilateral $ABCD$ be a parallelogram. Then

$$\overrightarrow{AB} = \overrightarrow{DC} \Rightarrow \overrightarrow{OB} - \overrightarrow{OA} = \overrightarrow{OC} - \overrightarrow{OD} \Rightarrow \vec{b} - \vec{a} = \vec{c} - \vec{d} \Rightarrow \vec{b} + \vec{d} = \vec{a} + \vec{c}$$

and hence $$\frac{\vec{b}+\vec{d}}{2} = \frac{\vec{a}+\vec{c}}{2}.$$

This shows that the position vectors of the midpoint of the line segments AC and BD are the same. In other words, the diagonals bisect each other.

Conversely let us assume that the diagonal bisects each other. Thus the position vectors of the midpoint of the line segments AC and BD are the same. Thus

$$\frac{\vec{a}+\vec{c}}{2} = \frac{\vec{b}+\vec{d}}{2} \Rightarrow \vec{a}+\vec{c} = \vec{b}+\vec{d} \Rightarrow \vec{c}-\vec{d} = \vec{b}-\vec{a}.$$

This implies that $\overrightarrow{OC} - \overrightarrow{OD} = \overrightarrow{OB} - \overrightarrow{OA}$ and hence $\overrightarrow{DC} = \overrightarrow{AB}$. This shows that the lines AB and DC are parallel. From $\vec{a}+\vec{c} = \vec{b}+\vec{d}$ we see that $\vec{a}-\vec{d} = \vec{b}-\vec{c}$ which shows that the lines AD and BC are parallel. Hence $ABCD$ is a parallelogram.

EXERCISE 8.1

(1) Represent graphically the displacement of

(i) $45cm$ $30°$ north of east. (ii) $80km$, $60°$ south of west

(2) Prove that the relation R defined on the set V of all vectors by '$\vec{a}$ R $\vec{b}$ if $\vec{a} = \vec{b}$' is an equivalence relation on V.

(3) Let $\vec{a}$ and $\vec{b}$ be the position vectors of the points A and B. Prove that the position vectors of the points which trisects the line segment AB are $\frac{\vec{a}+2\vec{b}}{3}$ and $\frac{\vec{b}+2\vec{a}}{3}$.

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